Flo

Go with the flow of life.

🏆 ANZ Diversity Hackathon 2026 · 🚌 Amazon Women in AI/ML · ⚡ Built with Kiro — AWS Agentic IDE · 📐 Architecture Document

🏗 Architecture Overview

Flo is a **deployed, production-grade MVP** built on a six-layer AWS architecture. The system uses **Amazon Bedrock's multi-agent framework** as its intelligence core — ten specialised agents coordinated by a supervisor orchestrator. Built spec-first using **Kiro** (AWS's agentic IDE), a two-person team delivered this system within the hackathon timeframe.

10	12	33	100+
Bedrock Agents deployed	AWS services integrated	Backend services built	Automated test cases

📐 Diagram 1 — Layer Architecture

Flo's internal service decomposition across six layers — from the React PWA down to the AWS services. Reflects **33 backend services**, **10 Bedrock agent types**, and a DynamoDB single-table design with two GSIs.

FLO — COMPLETE LAYER ARCHITECTURE

LAYER 1 — CLIENT (BROWSER / MOBILE)

React 18 PWA · Vite · TypeScript | Pages: Landing · Login · Onboarding · Dashboard | 15+ Components
Mobile Calendar · Mobile Dashboard · Mobile Nav | WebSocket Client · Offline Sync · Cache | Design System · Accessibility

↓ HTTP / REST API

LAYER 2 — API GATEWAY & AUTH

Amazon API Gateway | Express.js API Server on AWS Lambda | Auth: /register · /login · /logout | Events: /categories · /allocation
Dashboard: /metrics · /thresholds · /suggestions | Routes: /conflicts · /preferences · /sync
Amazon Cognito (Session · Token Validation)

↓ Service Calls

LAYER 3 — BUSINESS LOGIC (33 SERVICES)

Auth & Access | Event Management | Time Tracking | Notifications | Conflict Detector | Conflict Resolver
Summaries (Generator · Scheduler · Event-Triggered) | Thresholds | Calendar Sync Registry | Data Management | Onboarding
Audit Logger | Credential Encryption | Backup Monitor | Recovery Manager | Change Detector

↓ AI Agent Calls

LAYER 4 — AWS BEDROCK AGENTS (10 AGENTS)

1	2	3	4	5
Primary Calendar	Personal Calendar	Document & Newsletter	App & Message	Activities & Events
6	**7**	**8**	**9**	**10**
Weather & Environment	Conflict Resolution	Weekly Summary	Activity Classification	Orchestrator

Core: Lambda Handler · Agent Registry · Tool Functions (calendar-tool · weather-tool · parser-tool) · Error Handling · Bedrock Agent Monitoring · Bedrock Guardrails

↓ Data Operations

LAYER 5 — DATA ACCESS

DynamoDB Client Wrapper | Single-Table Design (FamilyCalendar) | GSI1 — Date Range Queries | GSI2 — Status / Type Queries
Query Optimisation · Connection Pooling | DynamoDB Streams (Change Detection · Audit) | ElastiCache (Sessions · Agent State)

↓ AWS SDK

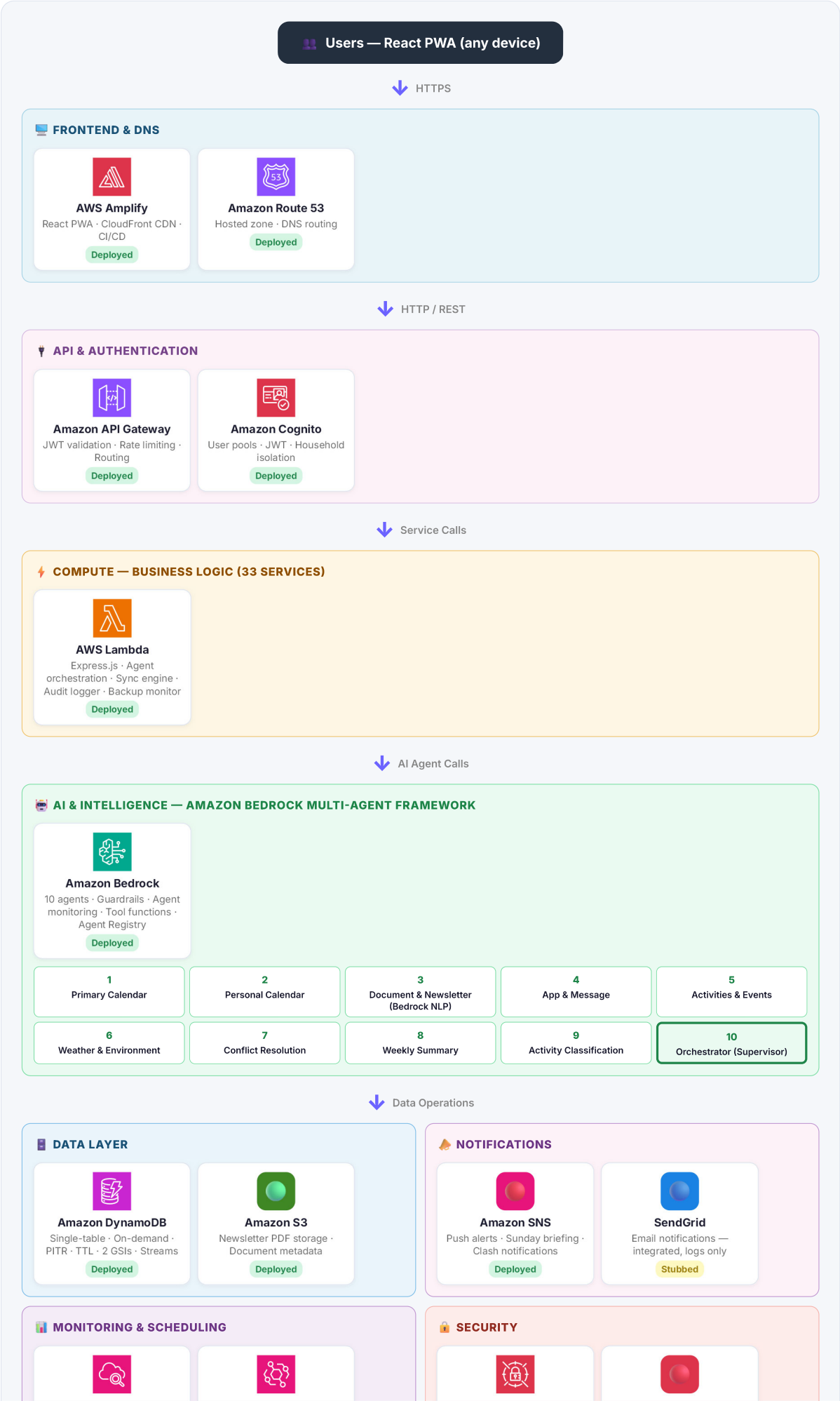
LAYER 6 — AWS SERVICES

Amazon DynamoDB ✅ | Amazon Bedrock ✅ | AWS Lambda ✅ | Amazon API Gateway ✅ | AWS Amplify ✅
Amazon Cognito ✅ | Amazon SNS ✅ | Amazon S3 ✅ | Amazon Route 53 ✅ | Amazon EventBridge ✅
AWS Secrets Manager ✅ | Amazon CloudWatch ✅ | SendGrid Email ⚠️ stubbed

Diagram 2 — AWS Service Architecture

End-to-end AWS service architecture using official AWS service icons. **Serverless-first** — Lambda for compute, DynamoDB on-demand, Bedrock for AI reasoning. Zero server management, auto-scaling built in.

Deployed · Stubbed · Referenced · OAuth backend built, UI in roadmap



Amazon CloudWatch
Metrics · Logs · Agent monitoring · Admin alerts
Deployed

Amazon EventBridge
Sunday briefing scheduler · Sync triggers · Cron jobs
Deployed

AWS Secrets Manager
4 secrets · OAuth credentials · API keys
Deployed

AWS KMS
Referenced in documentation
Referenced

EXTERNAL SOURCES — OAUTH BACKEND BUILT, UI CONNECTION IN ROADMAP

Google Calendar
OAuth service built
UI pending

Microsoft Outlook
OAuth service built
UI pending

Weather API
Live data · Integrated
Deployed

SeeSaw · SEQTA · Connect Now
School app parsing via Bedrock NLP
UI pending

Key Data Flow — School Communication Parsing

School communications are processed by Flo's Bedrock parser agents using **Bedrock NLP directly**. Raw content — email, HTML, PDF, or plain text — is passed to the agent which reasons over natural language to extract structured events. No structured date fields are required.

Implemented — Raw Content Parsing

SchoolNewsletterParserAgent and **UnstructuredDataParserAgent** accept raw content in any format (`email · html · pdf · text · json`) as a `rawData` input → **Bedrock Agent 3** reads the natural language content → foundation model extracts dates, deadlines, and action items → structured events returned → **Orchestrator Agent 10** cross-references with work and personal calendar data → clash detected → **Conflict Resolution Agent 7** drafts resolution → stored in **DynamoDB** → **Amazon SNS** notifies the user

Implemented — School App Messages (kids_school · kids_connect)

Bedrock Agent 4 (App & Message) reads teacher messages from SeeSaw, SEQTA, and Connect Now as text input → extracts embedded deadlines and action items from conversational natural language (e.g. *"Science project needs to come back by next Wednesday"*) → same downstream extraction and notification flow

Roadmap — Public URL Fetching

Most ANZ school newsletters are publicly accessible URLs on school websites. A future enhancement will allow parents to paste a newsletter URL → Lambda fetches the document directly → passes raw content to Bedrock Agent 3 → same extraction flow. This removes the need for manual upload entirely.

Roadmap — Direct School App API Integration

Direct API connections to SeeSaw, SEQTA, and Connect Now (Department of Education WA) will allow Flo to pull messages automatically without manual input. OAuth service pattern already established in the codebase.

Security & Privacy Architecture

Designed with privacy as a first principle. Aligned with the **Australian Privacy Act 1988** and the Notifiable Data Breaches Act 2017.

Data Isolation
Each household stored under unique Cognito user pool ID. No cross-household data access at any layer.

No Model Training
Bedrock never uses customer data to train foundation models. All data stays in the customer's AWS environment.

Secrets Manager
4 secrets deployed. OAuth credentials and API keys managed with secure rotation.

Bedrock Guardrails
Content filtering and PII detection applied to all model inputs and outputs across all 10 agents.

Audit Logging
All event changes and agent tasks logged with timestamp, user ID, and full context via DynamoDB Streams.

Minimal Retention
DynamoDB TTL auto-expires sessions (30d), tasks (7d), notifications (90d), audit logs (1yr).

🚀 Deployment Status

✅ FULLY DEPLOYED

✓ 10 Bedrock agents + Orchestrator
✓ Conflict detection & resolution engine
✓ Weekly summary generation
✓ Invisible Load Dashboard
✓ Weather & UV integration
✓ Cognito auth & household isolation
✓ Audit logging · backup monitor · credential encryption
✓ DynamoDB single-table · PITR · Streams
✓ Route 53 · Secrets Manager (4 secrets)
✓ 25+ frontend components · offline PWA
✓ 100+ automated test cases

🗺️ ROADMAP

◑ OAuth UI connection (backend 80% done)
◑ Real-time WebSocket dashboard
◑ SendGrid — end-to-end email delivery
◑ Voice briefing via Amazon Polly
◑ Carer-specific agent & GP integration
◑ Enterprise licensing via AWS Marketplace
◑ Mobile touch optimisation

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Sheryl Philip · Anita Birje | ANZ Diversity Hackathon — Amazon Women in AI/ML 2026 | Built with Kiro — AWS Agentic IDE